

Afra Feyza Akyürek

I focus on post-training, evaluation, and continual learning for large language models, with an emphasis on real-world, high-stakes reasoning. My recent work spans online reinforcement learning (OnlineRubrics, RL4F), agentic systems, human data programs for post-training and evaluation, and long-horizon workflows involving file processing and artifact generation. I led the development of PRBench and DrugDiscoveryBench, large-scale evaluations of language models and agents across professional domains including finance, legal services, healthcare, and life sciences.

EDUCATION

PhD in Computer Science, <i>Boston University</i>	September 2019 — March 2025
MSc in Statistics, <i>Carnegie Mellon University</i>	August 2018 — July 2019
BSc in Computer Engineering, <i>Koc University, Ranked 3rd in graduating class.</i>	September 2014 — July 2018

WORK EXPERIENCE

Research Scientist <i>Scale AI</i>	March 2025 — Present <i>San Francisco, CA</i>
- Research on LLM post-training (incl. online RL) and synthetic data generation for reasoning and agentic workflows. - Develop and evaluate agentic systems for healthcare, life sciences and professional domains, including artifact generation. - Lead the design and curation of PRBench, a large-scale benchmark for expert-level reasoning in finance and law, and DrugDiscoveryBench, a benchmark for coding agents in early-stage drug discovery. - Advise and shape internal human data programs for post-training and evaluation. - Analyze customer LLM capabilities and propose targeted data programs to address reasoning and deployment gaps.	
Research Intern <i>Allen Institute for Artificial Intelligence (AI2)</i>	September 2022 — December 2022 <i>Seattle, WA</i>
Designed and implemented an external critique model trained with reinforcement learning to repair errors in LLM outputs (ACL 2023).	
Machine Learning Research Intern <i>Apple</i>	June 2021 — August 2021 <i>Cambridge, MA</i>
Evaluating and aligning large language models for social biases (Findings of NAACL 2022).	
Statistical Learning Intern <i>Novartis</i>	May 2019 — July 2019 <i>East Hanover, NJ</i>
Developed machine learning model to predict CAR-T therapy outcomes, accurately forecasting treatment efficacy and safety risks; findings presented to VP leadership.	

PUBLICATIONS

- A.F.A, Xinming Tu, Alec Gutmanstein, et al. [DrugDiscoveryBench: Can Coding Agents Assist Early-Stage Drug Discovery?](#) Preprint, 2026
— *Evaluation of coding agents on multi-step computational and information-retrieval workflows in early-stage drug discovery.*
- A.F.A, Advait Gosai, Chen Bo Calvin Zhang, et al. [PRBench: Large-Scale Expert Rubrics for Evaluating High-Stakes Professional Reasoning](#). To appear in ACL 2026
— *Large-scale evaluation of LLMs for reasoning in professional domains.*
- MohammadHossein Rezaei, Robert Vacareanu, Zihao Wang, Clinton Wang, Bing Liu, Yunzhong He, A.F.A. [Online Rubrics Elicitation from Pairwise Comparisons](#). ICML, 2026
— *Online estimation of rewards via synthetic rubrics.*
- A.F.A, Ekin Akyürek, Leshem Choshen, Derry Wijaya, Jacob Andreas. [Deductive Closure Training of Language Models for Coherence, Accuracy, and Updatability](#). Findings of ACL, 2024
— *Built a self-training algorithm that makes language models significantly more accurate and coherent on factual knowledge.*
- A.F.A, Eric Pan, Garry Kuwanto, Derry Wijaya. [DUnE: Dataset for Unified Editing](#). EMNLP, 2023
— *Designed a new benchmark that tests targeted model editing based on natural language instructions and preferences. I showed surgical model editing methods generalize poorly compared to simple RAG systems.*

- **A.F.A**, Ekin Akyürek, Ashwin Kalyan, Peter Clark, Derry Wijaya, Niket Tandon. [RL4F: Generating Natural Language Feedback with Reinforcement Learning for Repairing Model Outputs](#). **ACL**, 2023
— Demonstrated that a small model (weak) can be trained via RL to give language feedback to large model (strong) during test-time yielding to improvements in strong models accuracy — an early demonstration of how a two-agent system of LLMs could operate.
- **A.F.A**, Muhammed Yusuf Kocyigit, Sejin Paik, and Derry Wijaya. [Challenges in Measuring Bias via Open-Ended Language Generation](#). **GeBNLP at NAACL (Oral)**, 2022
— Identified the pitfalls in evaluating social biases in language model generations and recommended a robust evaluation scheme.
- Garry Kuwanto*, **A.F.A***, Isidora Chara Tourni*, Siyang Li*, Alex Jones, Derry Wijaya. [Low-Resource Machine Translation Training Curriculum Fit for Low-Resource Languages](#). **PRICAI**, 2023
- **A.F.A**, Sejin Paik, Muhammed Yusuf Kocyigit, Seda Akbiyik, Şerife Leman Runyun, and Derry Wijaya. [On Measuring Biases in Prompt-Based Learning](#). **Findings of NAACL**, 2022
- **A.F.A**, Ekin Akyürek, Derry Wijaya and Jacob Andreas. [Subspace Regularizers for Few-Shot Class Incremental Learning](#). **ICLR**, 2022
— Made image classifiers learn continually using a small set of examples and language instructions without much forgetting by a simple regularized objective that achieved SoTA.
- Ekin Akyürek, **A.F.A** and Jacob Andreas. [Learning to Recombine and Resample Data for Compositional Generalization](#). **ICLR**, 2021
- Haryo Akbarianto Wibowo, Made Nindyatama Nityasya, **A.F.A**, Suci Fitriany, Alham Fikri Aji, Radityo Eko Prasajo, and Derry Tanti Wijaya. [IndoCollex: A Testbed for Morphological Transformation of Indonesian Colloquial Words](#). **Findings ACL-IJCNLP**, 2021
- **A.F.A**, Lei Guo, Randa Elanwar, Margrit Betke, Prakash Ishwar and Derry T. Wijaya. [Multi-label and Multilingual News Framing Analysis](#). **ACL**, 2020
— Built a SoTA classifier to identify the ways different media outlets frame the same events across multiple languages.

LEADERSHIP, HONORS AND AWARDS

Hariri Institute for Computing , Graduate Student Fellow	May 2021 - December 2024
Best Senior Design , Self-Driving Cars in Unity with Deep Reinforcement Learning	June 2018
Hult Prize , Regional Finalist	March 2017
Society of Women Engineers , President, Koc University	June 2016 - May 2017
31st Place in National University Entrance Exam amid 2 million takers in Turkey	2013